

Instructions:

1. Write the Code:

- Implement your solution in code. Follow best practices for coding standards, such as meaningful variable names, proper indentation, and comments where necessary.
- Break down the problem into smaller functions or modules to improve code readability and maintainability.

2. Implement the wireFrame

- Screenshots are for reference only you can build your own user interface.
- Using any framework like angular,reactjs,etc will be a plus point.
- Use mock data to implement the feature if api is not available.

3. Document Your Code :

- Consider adding documentation or comments to explain the logic and reusable parts, especially for complex parts or algorithms.

4. Submit Your Solution:

- Once you're satisfied with your code and it meets all the requirements, submit your solution on GitHub and share the GitHub link.

5. Demonstration:

- Include a demonstration video showcasing key features of the ride-sharing platform. - Alternatively, use screenshots to visually highlight the user interface and functionality.

Plus Points in Implementation (Overall Evaluation Criteria):

1.Complexity Estimation:

- Conduct a thorough analysis of time and space complexity in the system.
- Utilize efficient algorithms and data structures to optimize both time and space requirements.
- Regularly monitor and analyze resource consumption for cost-effective scaling.

2. User Experience:

- Is the user interface intuitive and easy to navigate?

- Does the system provide feedback to users regarding successful or unsuccessful operations?

3. Error Handling:

- Does the system handle errors gracefully, providing informative error messages to users?
- Are there checks in place to prevent invalid operations or data inconsistencies?

4. Performance:

- How efficiently does the project perform its tasks, especially when handling large datasets or high loads?
- Is there any noticeable lag or delay in response times, especially when handling a large number of operations?

5. Scalability:

- Can the system accommodate an increasing number of users without significant performance degradation?
- Is the system designed to be easily scalable and adaptable to changing requirements or business needs?

6. Functionality:

- Does the project fulfill its intended purpose effectively?
- Are all essential features and requirements implemented?
- Is the project user-friendly and intuitive to use?

1. Visitor Management System

A **Visitor Management System (VMS)** plays a crucial role in ensuring workplace security, regulatory compliance, and a seamless visitor experience. This system efficiently tracks and manages visitors while maintaining strict access control. Below is a detailed breakdown of its functionalities:

I. Visitor Registration

When a visitor arrives at the workplace, their details are captured systematically to ensure security and record-keeping.

✓ Information Captured:

- **Full Name** – The visitor's legal name for identification.
- **Contact Information** – Mobile number and/or email for communication and verification.

- **Purpose of Visit** – The reason for their visit (e.g., meeting with an employee, maintenance work, interview, delivery, etc.).
- **Host Employee Details** – Name and department of the employee they are visiting.
- **Company/Organization Name** – If the visitor represents a business, their company details are recorded.
- **Check-in and Check-out Time** – Automatic logging of entry and exit times for security tracking.
- **Mandatory Photo Capture** – A photograph is taken at the registration desk for identity verification.

◆ **Process:**

1. The **security guard** or **self-service kiosk** collects visitor details and captures their photo.
 2. The system automatically sends a request to the respective host employee for approval.
 3. A visitor badge (physical or digital QR code) is generated after approval.
-

II. Approval Workflow

To prevent unauthorized access, the system ensures that every visitor must be approved by the host employee.

✅ **Approval Steps:**

1. **Real-Time Notification:** The host employee receives an instant alert (email, SMS, or app notification) regarding the visitor request.
2. **Quick Approval/Rejection:** The employee can approve or deny the visitor's request through the mobile app, web portal, or an automated IVR system.
3. **Access Grant:** Upon approval, the visitor is allowed entry, and a digital or printed visitor pass is issued.
4. **Denied Access:** If rejected, the visitor is not permitted inside, and security is notified to handle the situation accordingly.

◆ **Security Benefits:**

- Ensures that only authorized visitors can enter the premises.
 - Reduces manual intervention and enhances operational efficiency.
 - Tracks approval history for audit purposes.
-

III. Pre-Approval for Convenience

To minimize last-minute approval delays, employees can pre-approve visitor requests.

✓ Features of Pre-Approval:

- Employees can schedule and approve visitor access in advance for a specific **date and time window** (e.g., 10 AM - 12 PM).
- Pre-approved visitors receive a **QR code/e-pass** via email or SMS, which they can scan upon arrival to bypass the manual approval process.
- If a visitor fails to check in within the approved window, the request **expires automatically** for security purposes.
- Admins can enforce rules, such as **pre-approval limits** (e.g., max 5 visitors per employee per day).

◆ Use Cases:

- Scheduling **client meetings** or **vendor visits** in advance.
- Fast-tracking **frequent visitors**, such as maintenance personnel.
- Reducing **employee workload** on the day of the visit.

IV. Screenshots:

Invite a visitor:

The screenshot shows a web application interface for inviting visitors. At the top, there is a header bar with a back arrow and the text 'Invite Visitors'. Below this, the form is divided into several sections. On the left, there is a calendar icon and a label 'Event Title *' followed by a text input field. To the right of this is a search bar with a magnifying glass icon and the placeholder text 'Search by name, id, email or phone *'. Below the 'Event Title' field, there is a dropdown menu labeled 'Types of Visit *' with 'Business Guests' selected. A list of options is visible: Business Guests, Vendor, Personnel, Govt Officials, Interview, PwC Network Firm, and Others. To the right of the 'Types of Visit' dropdown is another dropdown labeled 'Office *' with 'Mumbai Goregaon' selected. Below these dropdowns, there is a section for 'Optional' information, which includes a text input field and a checkbox. At the bottom of the form, there is a button labeled 'Confirm Invite'.

Search and add employee:

← Invite Visitors

Event Title *

Meeting

Types of Visit *

Business Guests

Office *

Mumbai Goregaon

Feb 10 2025

07:00 pm

-

08:00 pm

Search by name, id, email or phone *

ADDED GUESTS

TN

Teja neeradi

×

AT

Akshay Test

×

Personal note to guests

Optional

Please report at Front Desk

Confirm Invite

Front desk application:

Visitors

All (27)

Search by name,email or phone

Feb 10 2025

12:00 am - 11:59 pm

Visitor	Type of Invite	Entry Time	Exit Time	Status
Dhulabhai Bamania Host: Lalita Mehta	Contract Staff Self Check-in	04:21 pm	12:21 am	OVERSTAY
Ajay Singh Host: Lalita Mehta	Contract Staff Self Check-in	08:04 pm	04:04 am	SELF CHECK-OUT
Arun Kumar Host: Lalita Mehta	Contract Staff Self Check-in	11:46 pm	07:46 am	SELF CHECK-OUT
Navin Patidar Host: Lalita Mehta	Vendor Self Check-in	06:43 am	03:43 pm	OVERSTAY
Sukhdev Ahari Host: Lalita Mehta	Contract Staff Self Check-in	07:10 am	04:10 pm	SELF CHECK-OUT
Vinod Dindor Host: Lalita Mehta	Contract Staff Self Check-in	07:14 pm	04:14 pm	SELF CHECK-OUT
Geetagothil Host: Lalita Mehta	Contract Staff Self Check-in	07:38 am	04:38 pm	OVERSTAY
Jayesh Singh Host: Lalita Mehta	Contract Staff Self Check-in	07:40 am	03:40 pm	SELF CHECK-OUT
Varun Maru Host: Lalita Mehta	Contract Staff Self Check-in	07:57 am	04:57 pm	SELF CHECK-OUT
Nilesh Pandey Host: Lalita Mehta	Contract Staff Self Check-in	08:57 am	05:57 pm	OVERSTAY
Sanjay Gandhi	Contract Staff	09:02 am	06:02 pm	OVERSTAY

Guest Details

OVERSTAY

DB

Dhulabhai Bam...

8980904195

Guest

LM

Lalita Mehta

lalita.mehta@pwc.com

9756195792

Host

Check-In

4:21 PM

Check-Out

12:21 AM

Walk In Visitor - Dhulabhai bamania

Sun, Feb 9, 04:21 PM - Mon, Feb 10, 12:21 AM

Contract Staff

Other Details

Company Name : Walsons

Role : Security

Sponsor LOS : IFS

Temp Card No : 16

Additional Information

Input text

> 0/1000

Check-Out

2. Vendor Cab and Driver Onboarding & Vendor Hierarchy Management:

A **Vendor Cab and Driver Onboarding System** enables seamless **multi-level vendor management, vehicle onboarding, document verification, and hierarchical access control**. This system ensures that **Super Vendors and Sub Vendors** can efficiently manage fleets, onboard drivers, and maintain compliance, while providing structured delegation capabilities.

I. Multi-Level Vendor Hierarchy

To support **large-scale fleet operations**, the system allows vendors to operate at **multiple hierarchical levels**, ensuring **structured access control and management**.

✓ Flexible N-Level Hierarchy

- Vendors can operate at **multiple levels**, forming a **parent-child relationship** (e.g., Super Vendor → Regional Vendor → City Vendor → Local Vendor).
- This hierarchy ensures that **fleet management responsibilities are distributed**, preventing operational bottlenecks.

✓ Role-Based Access Management

- Each vendor level has **specific roles and permissions**, ensuring **controlled access to vehicles, drivers, and fleet operations**.
- **Super Vendors** can **define access policies** for their sub-vendors.

♦ Example Use Case:

- A **National Fleet Operator (Super Vendor)** has **Regional Vendors**, who in turn manage **City-Level Vendors** handling **local cabs and drivers**.

♦ Outcome:

- Ensures **scalability** for large vendor networks.
- **Prevents unauthorized access** by implementing a structured hierarchy.

II. Super Vendor Access & Delegation

Super Vendors have complete control over **sub-vendors, fleets, and driver onboarding**, ensuring centralized management.

✓ Access Delegation to Sub Vendors

- **Super Vendors** can **grant specific access permissions** to sub-vendors to manage:
 - **Fleet onboarding & assignments**
 - **Driver onboarding & verification**
 - **Operational tasks (booking management, payments, compliance tracking, etc.)**

✓ Delegation of Authority

- A **Super Vendor** can **authorize a sub-vendor** to act on their behalf for certain operations.
- The **delegated sub-vendor** can perform **all or selected** administrative actions under the Super Vendor's name.

✓ Controlled Delegation Rights

- The **Super Vendor** can:
 - **Enable or revoke delegation at any time.**
 - Restrict access to **specific functions** (e.g., sub-vendor can onboard drivers but not process payments).

♦ Example Use Case:

- A **Super Vendor managing a large fleet** can delegate responsibilities to a **Regional Vendor**, who further assigns tasks to **City-Level Vendors**.
- The Super Vendor ensures **smooth workflow automation** without handling **every small task manually**.

♦ Outcome:

- **Reduces administrative overhead** for Super Vendors.
- **Ensures continuity** of operations even if the Super Vendor is unavailable.

III. Sub-Vendor Fleet & Driver Management

Sub Vendors are responsible for **managing vehicle onboarding, driver assignments, and compliance documentation**.

✓ Vehicle & Driver Onboarding

- Sub Vendors can onboard:
 - **Cabs & Commercial Vehicles** – Enter vehicle details such as **registration number, model, seating capacity, fuel type, etc.**
 - **Drivers** – Add drivers and assign them to specific vehicles.

✓ Driver Document Upload

- Sub Vendors can upload **essential driver documents**:
 - **Driving License (DL)**
 - **Vehicle Registration Certificate (RC)**
 - **Permit & Pollution Certificate.**

♦ Example Use Case:

- A **City-Level Vendor** manages **local fleet onboarding and assigns drivers** to cabs.
- The system **flags expired documents** and prevents non-compliant vehicles from operating.

♦ Outcome:

- **Ensures regulatory compliance** with up-to-date documentation.
- **Reduces manual document tracking efforts** with automated reminders.

IV. Super Vendor's Complete Control Over Sub-Vendors

Super Vendors have **full visibility and control** over their **entire sub-vendor network**, including **fleet status, driver assignments, and compliance reports**.

✓ Centralized Dashboard for Super Vendors

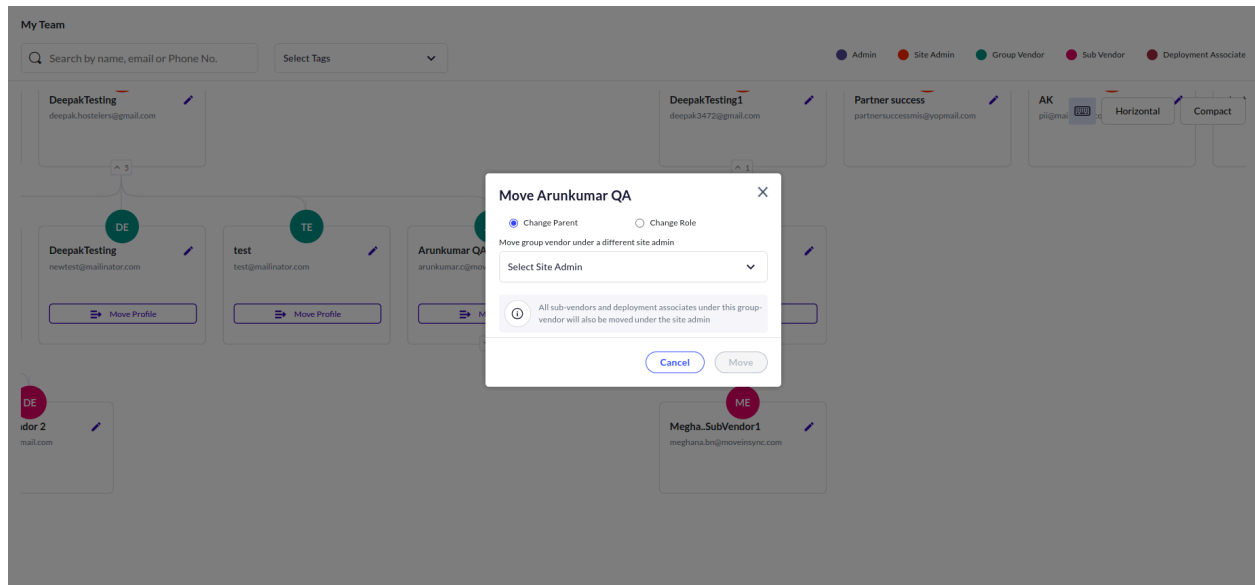
- A **single dashboard** provides:
 - **Real-time view** of all sub-vendors.
 - **Fleet status updates** (number of active/inactive vehicles).
 - **Pending document verifications & approvals.**
 - **Driver availability tracking.**

✓ Super Vendor Action Control

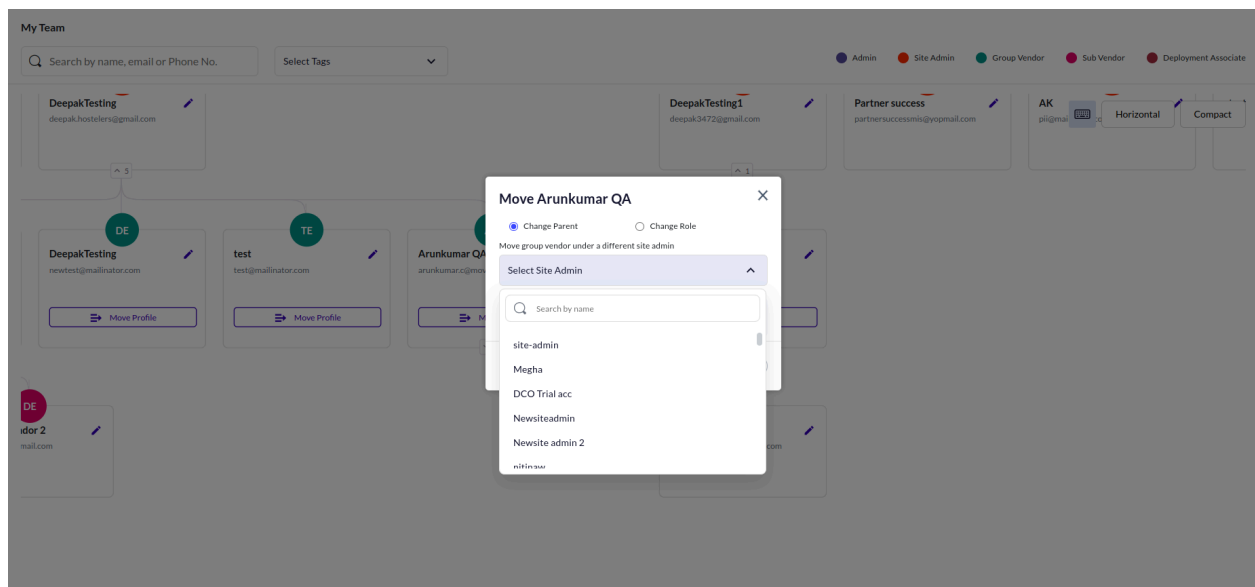
- The **Super Vendor can override sub-vendor actions**, ensuring compliance and smooth operations.
- **System-wide reports & analytics** help **optimize fleet performance**.

♦ Example Use Case:

Move lower hierarchy user to other manager:



Search manager and move:



3. Shuttle Management System: A Smart Campus Transit Solution.

The **Shuttle Management System** is designed to provide **efficient, cost-effective, and seamless** transportation for students within a university campus. The system streamlines shuttle bookings, trip tracking, and driver availability management, ensuring a hassle-free commuting experience.

I. Shuttle Booking Management

- **Add, View, and Edit Bookings:**
 - Users (students or staff) can book shuttle rides through the system.
 - Admins can view and modify existing bookings if needed.
- **Trip History Tracking:**
 - Users can view their past trips, including details like date, time, route, and driver information.

II. Driver Availability Management

- **Driver Schedule Timeline View:**
 - A visual timeline displaying the availability of drivers for a given day.
 - Ensures admins can efficiently allocate drivers based on demand.
- **Hourly Scheduling Controls:**
 - Admins can add, view, and modify a driver's **start duty, end duty, and breaks** within an **hourly time frame**.
 - Helps optimize driver shifts and reduces idle time.

III. Admin Functionalities

- **Manage Shuttle Routes:**
 - Define and update shuttle routes, including pickup/drop-off points.
- **Assign Drivers to Routes:**
 - Ensure that each shuttle trip has an available driver.
- **Monitor Shuttle Usage & Demand:**
 - Track peak hours and optimize shuttle schedules accordingly.

IV. Benefits & Impact

- **Enhanced Convenience:** Students can seamlessly book rides and track shuttle availability.
- **Optimized Driver Scheduling:** Reduces idle time and ensures efficient workforce management.

- **Cost-Effective Operations:** Balances demand with available drivers, reducing unnecessary trips.

Screenshots

Driver journey view and edit:

Driver Management

Dec 16, 2021

Search driver

6:007:008:009:0010:0011:0012:0013:0014:0015:0016:0017:0018:0019:0020:0021:0022:00

Samuel Jones
Online

Bob Jones
Offline

Jonathan Spikes
Offline

→ Start Duty

→ End Duty

☰ Add Break

2 Pickup

3 Drop

→ Duty Start

→ Duty End

📍 Pickup/Drop

☰ Break

🚗 Vehicle Change

→ Empty Leg

Booking Management

Dec 16, 2024

Search Emp, ID, Booking ID

Booking ID	Employee	Status	From	To	Vehicle	Requested Pickup Time	Pickup Time	Planned Drop	Actual Drop	
123123	Thompson	Accepted	Library	Data Centre	NB-002-RF	11:21	-	11:32	11:32	View
324235	Daniel Radcliff	Waiting	Library	Parking	NB-002-RF	11:34	-	11:43	11:43	View
545232	W.J. Smith	No Show	Data Centre	Parking	NB-002-RF	11:50	-	12:10	-	View
434532	Tina Shah	Declined	Library	Data Centre	-	11:55	-	-	-	View
545232	W.J. Smith	Completed	Data Centre	Parking	NB-002-RF	11:58	12:25	12:35	12:40	View
434532	Tina Shah	Requested	Library	Data Centre	NB-002-RF	11:55	-	12:15	-	View
545232	W.J. Smith	On Going	Data Centre	Parking	-	11:58	12:25	12:35	-	View
434532	Tina Shah	Cancelled	Library	Data Centre	-	11:55	-	12:15	-	View
545232	W.J. Smith	Dropped	Data Centre	Parking	NB-002-RF	11:58	12:25	12:35	12:40	View
434532	Tina Shah	Declined	Library	Data Centre	-	11:55	-	-	-	View

Showing 1-10 of 50 items

1

2

...

5

>

[illegible]

Booking Management								Q Search
Booking ID	Employee	Status	From	To	Vehicle	Requested Pickup Time	Pickup Time	
123123	Thompson	Accepted	Library	Data Centre	NB-002-RF	11:21	-	
324235	Daniel Radcliff	Waiting	Library	Parking	NB-002-RF	11:34	-	
545232	W.J. Smith	No Show	Data Centre	Parking	NB-002-RF	11:50	-	
434532	Tina Shah	Declined	Library	Data Centre	-	11:55	-	
545232	W.J. Smith	Completed	Data Centre	Parking	NB-002-RF	11:58	12:25	
434532	Tina Shah	Cancelled	Library	Data Centre	NB-002-RF	11:55	-	